Tutorial 1: FINA2205
Review of Financial Statistics

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Course overview & logistics

Unit Structure

There are 11 lecture weeks (weeks 1-6, 8-12) and 8 tutorials for this unit. In Week 7, an online mid-semester exam is due on Thursday, 16 April 2025, and there are no lectures or tutorials that week.

Required Assessments

Best 5 out of 8

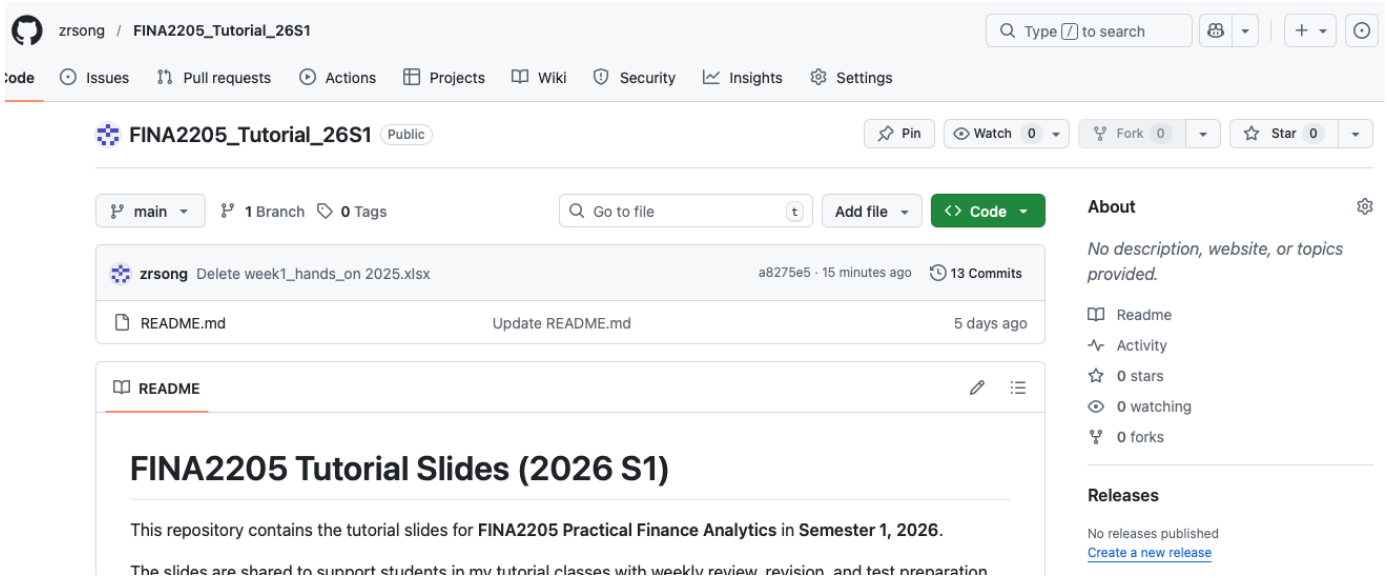
Nominate a leader & notify me by the end of tutorial 2 (4-5 per group)

Assessment #	Assessment Task	Weight	Assessment period/date/s	Submission Procedure	Unit Learning Outcome	Waiver
1	Participation	10%	10/3 12:00 AM - 15/5 11:59 PM	TBA	1 - 7	No
2	Group project	30%	04/5 11:59 PM	via LMS	1 - 7	No
3	Mid-Semester examination	10%	15/4 08:00 AM - 16/4 11:59 PM	via LMS	1 - 6	No
	Final examination	50%	31/5 09:00 AM - 14/6 04:00 PM	Check Student Connect	1 - 6	No

Source: https://alt-5ddb108fe0c42.blackboard.com/bbcswebdav/institution/Unit_Outlines_2026/FINA2205_SEM-1_2026/FINA2205_SEM-1_2026_UnitOutline.html?one_hash=93136EA055E02F072EBDEDD19B6378D7&f_hash=F5BA865BF7C0C09299478B94D2D20BE2

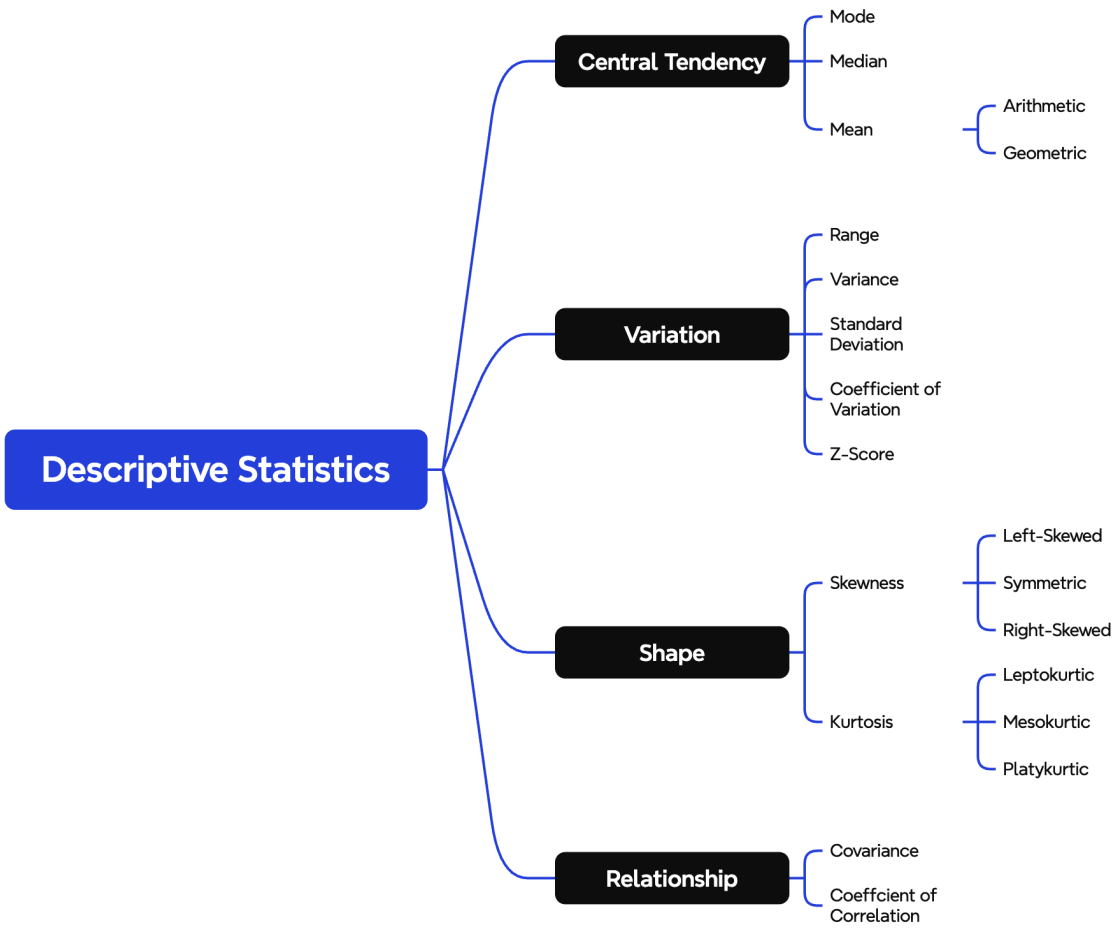
Where to find the slides

- I will upload our tutorial slides every **Friday** via my Github's repo.
- https://github.com/zrsong/FINA2205_Tutorial_26S1/tree/main



Agenda

- 1a (5 problems)
- 1b (5 problems)
- Additional questions (4 problems)



Presented with xmind

Recap: central tendency

- It means the typical or middle value of a set of data.
- It answers the question: *“What number best represents the whole group?”*

Mode

- The mode is the value that appears **most frequently** in a dataset.

Mean vs. Median

- Mean is the **average** value of a group of numbers.
- Median is the **middle value** after arranging the numbers from smallest to largest.

Recap: central tendency

Example 1

- Numbers: 6, 100, 8, 2, 4
- $Mean = \frac{2+4+6+8+100}{5} = 24$
- 2, 4, 6, 8, 100 $\longrightarrow$ $Median = 6$

How about the median of the dataset: 2, 4, 6, 8, **50,** 100?

$$Median = \frac{6+8}{2} = 7$$

Rule: when there is an even number of observations, the median is the average of the two middle values.

Recap: central tendency

Arithmetic mean vs. geometric mean

- Arithmetic mean = the usual average
  - Treats the numbers as **separate values**
  - Used for ordinary values such as test scores, salaries, or prices
- Geometric mean = the average rate of change over time
  - Treats the numbers as **connected changes** over time
  - Common examples: investment returns, growth rates, inflation rates

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

For the arithmetic mean

$$\overline{R}_G = [(1+R_1)\times(1+R_2)\times\cdots\times(1+R_n)]^{1/n} - 1$$

For the geometric mean

Recap: central tendency

An investment of \$100,000 declined to \$50,000 at the end of year one and rebounded to \$100,000 at end of year two:

$$X_1 = \$100{,}000 \quad X_2 = \$50{,}000 \quad X_3 = \$100{,}000$$

50% decrease 100% increase

The overall two-year return is zero, since it started and ended at the same level.

$$\overline{X} = \frac{(-.5)+(1)}{2} = .25 = 25\%$$

Misleading result

$$\overline{R}_G = [(1+R_1)\times(1+R_2)\times\cdots\times(1+R_n)]^{1/n} - 1$$
$$= [(1+(-.5))\times(1+(1))]^{1/2} - 1$$
$$= [(.50)\times(2)]^{1/2} - 1 = 1^{1/2} - 1 = 0\%$$

Sensible result

Rule: The arithmetic mean is always greater than or equal to the geometric mean, as arithmetic ignores the **"compounding"** effect.

Recap: central tendency

Financial application: geometric mean of the rate of return

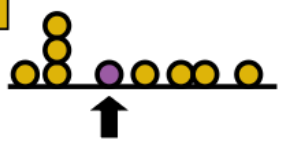
$$\bar{R}_G = [(1+R_1)\times(1+R_2)\times\cdots\times(1+R_n)]^{1/n} - 1$$

Central Tendency

- **Arithmetic Mean**

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

- **Median**



Middle value in the ordered array

- **Mode**

Most frequently observed value

- **Geometric Mean**

$$\bar{X}_G = (X_1 \times X_2 \times \cdots \times X_n)^{1/n}$$

Rate of change of a variable over time

1a-1

1. A survey reported the results for the salaries of professors teaching statistics in research universities with four to five years in the rank of associate professor and professor in the accompanying table. Interpret the median salary for the associate professors and professors.

Title	Median
Associate professor	$82,400
Professor	$108,600

Interpret the median salary for the associate professors. Choose the correct answer below.

- **A.** The arithmetic average of the associate professors' salaries is $82,400.
- **B.** The most common value for an associate professor's salary is $82,400.
- **C.** Half of the salaries for associate professors are smaller than or equal to $82,400, and the other half of the salaries for associate professors are larger than or equal to $82,400.
- **D.** Half of the salaries for associate professors are smaller than or equal to $108,600, and the other half of the salaries for associate professors are larger than or equal to $108,600.

Interpret the median salary for the professors. Choose the correct answer below.

- **A.** The arithmetic average of the professors' salaries is $108,600.
- **B.** Half of the salaries for professors are smaller than or equal to $108,600, and the other half of the salaries for professors are larger than or equal to $108,600.
- **C.** The most common value for a professor's salary is $108,600.
- **D.** Half of the salaries for professors are smaller than or equal to $82,400, and the other half of the salaries for professors are larger than or equal to $82,400.

1a-1 solution

- **C.** Half of the salaries for associate professors are smaller than or equal to $82,400, and the other half of the salaries for associate professors are larger than or equal to $82,400.

- **B.** Half of the salaries for professors are smaller than or equal to $108,600, and the other half of the salaries for professors are larger than or equal to $108,600.

1a-2

2. Suppose the rate of return for a particular stock during the past two years was 10% and 35%. Compute the geometric mean rate of return. (Note: A rate of return of 10% is recorded as 0.10, and a rate of return of 35% is recorded as 0.35.)

The geometric mean rate of return is _____%.
(Round to one decimal place as needed.)

1a-2 solution

$$\overline{R}_G = [(1+R_1)\times(1+R_2)\times\cdots\times(1+R_n)]^{1/n} - 1$$

$$\bar{R} = [(1+0.1)(1+0.35)]^{1/2} - 1$$
$$\bar{R} = 0.2186 = 21.9\%$$

1a-4

4. A company's stock price rose 361.4% in 2004, but dropped 78.0% in 2005.

a. Compute the geometric mean rate of return for the two-year period 2004 – 2005. (Hint: Denote a decrease of 78.0% by – 0.78.)
b. If someone purchased $1,000 of the company's stock at the start of 2004, what was its value at the end of 2005?
c. In 2006, the company's stock increased by 9.7%. Compute the geometric mean rate of return for the three-year period 2004 – 2006.

a. The geometric mean rate of return per year for the two-year period 2004 – 2005 was _____%.
(Type an integer or decimal rounded to two decimal places as needed.)
b. If someone purchased $1,000 of the company's stock at the start of 2004, its value at the end of 2005 was $_____.
(Round to the nearest cent as needed.)
c. The geometric mean rate of return for the three-year period 2004 – 2006 was _____%.
(Type an integer or decimal rounded to two decimal places as needed.)

1a-4 solution

a.

$$\bar{R}_G = [(1+R_1)\times(1+R_2)\times\cdots\times(1+R_n)]^{1/n} - 1$$

$$\bar{R} = [(1+3.614)(1-0.78)]^{1/2} - 1$$
$$\bar{R} = 0.0075 = 0.75\%$$

b. If someone purchased $1,000 of the company's stock at the start of 2004, its value at the end of 2005 is $1000(1.0075) = $1015.18.

c.

$$\bar{R} = [(1+3.614)(1-0.78)(1+0.097)]^{1/3} - 1$$
$$\bar{R} = 0.0365 = 3.65\%$$

Recap: measures of variation

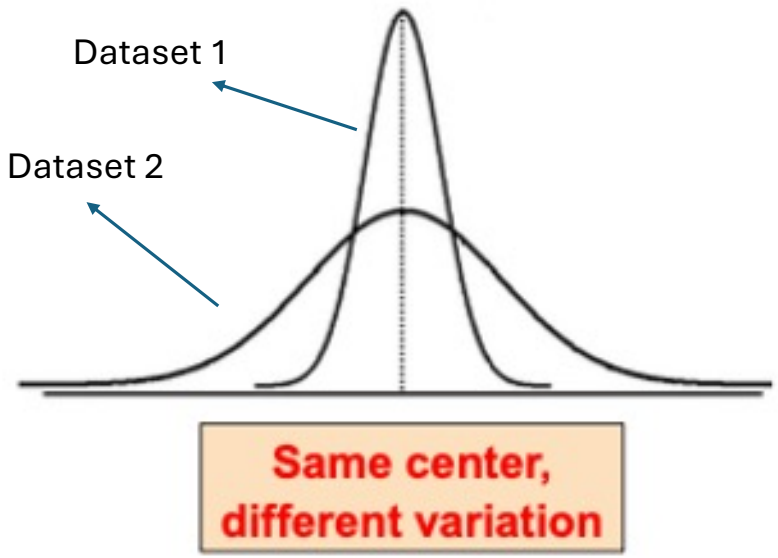
Example

- Dataset 1: 68, 69, 70, 71, 72
- Dataset 2: 40, 55, 70, 85, 100

$$mean = \frac{68 + 69 + 70 + 71 + 72}{5}$$

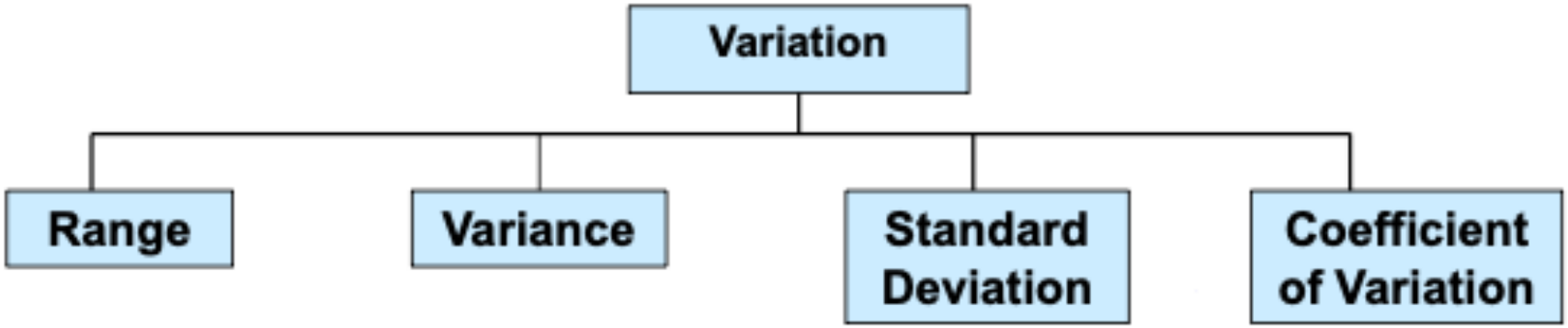
$$= \frac{40 + 55 + 70 + 85 + 100}{5} = 70$$

- Dataset 1 is much more concentrated, while dataset 2 is much more spread out.



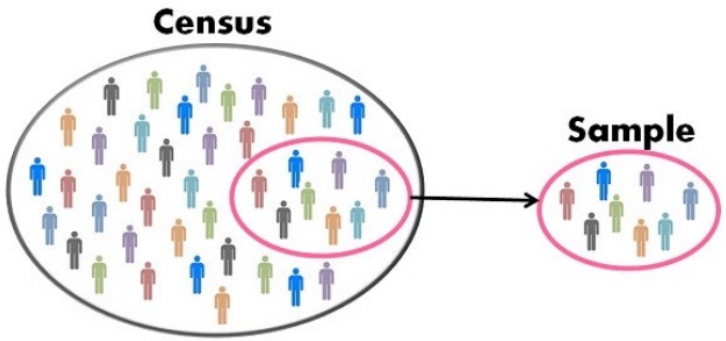
Recap: measures of variation

- It shows how spread out the data are.
- It answers the question: *"Are the values close together or far apart?"*
- Range: the difference between the largest and smallest values
  - Gives a quick idea of the spread



Recap: measures of variation

- Variance (S^2/σ^2): measures how far the values are from the mean
  - **Rule**: a larger variance means the data are more spread out
- Standard deviation (also called Volatility) (S/σ): is derived from variance and is equal to its square root.



$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}; S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

For a sample

$$\sigma^2 = \frac{\sum_{i=1}^{n}(X_i - \mu)^2}{n}; \sigma = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \mu)^2}{n}}$$

For a population

Recap: measures of variation

- The population variance vs. the sample variance
- Population variance
  - 100% information: we have data for every single member
  - The mean is exact, so the calculated spread is perfectly accurate
  - Just divide by the total number, n
- Sample variance
  - We only have a subset of the data
  - Samples naturally cluster around their own mean, making us underestimate the true volatility
  - Divide by a smaller number $(n - 1)$ artificially – **"degrees of freedom"**

1b-3

3. The last four years of returns for a stock are as shown here: [3].
   a. What is the average annual return?
   b. What is the variance of the stock's returns?
   c. What is the standard deviation of the stock's returns?
   Note: Notice that the average return and standard deviation must be entered in percentage format. The variance must be entered in decimal format.

Review Only

Click the icon to see the Worked Solution (Formula Solution).

a. What is the average annual return?

The average return is _____%. (Round to two decimal places.)

b. What is the variance of the stock's returns?

The variance of the returns is _____. (Round to five decimal places.)

c. What is the standard deviation of the stock's returns?

The standard deviation is _____%. (Round to two decimal places.)

3: Data Table

(*Click on the icon located on the top-right corner of the data table below in order to copy its contents into a spreadsheet.*)

Year	**1**	**2**	**3**	**4**
Return	− 4.4%	+ 28.4%	+ 11.7%	+ 3.9%

1b-3 solution

a.

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

- $Mean = \frac{-0.044+\cdots+0.039}{4} = 0.099 = 9.90\%$

b.

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}; S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

- $S^2 = \frac{(-0.044-0.099)^2+\cdots+(0.039-0.099)^2}{4-1} = 0.01953$
- $S = \sqrt{0.01953} = 0.1398 = 13.98\%$

1b-4

4. The last four years of returns for a stock are as shown here: [4].
   a. What is the average annual return?
   b. What is the variance of the stock's returns?
   c. What is the standard deviation of the stock's returns?
   Note: Notice that the average return and standard deviation must be entered in percentage format. The variance must be entered in decimal format.

Review Only

Click the icon to see the Worked Solution (Formula Solution).

a. What is the average annual return?

The average return is _____%. (Round to two decimal places.)

b. What is the variance of the stock's returns?

The variance of the returns is _____. (Round to five decimal places.)

c. What is the standard deviation of the stock's returns?

The standard deviation is _____%. (Round to two decimal places.)

4: Data Table

(*Click on the icon located on the top-right corner of the data table below in order to copy its contents into a spreadsheet.*)

Year	**1**	**2**	**3**	**4**
Return	− 4.3%	+ 28.4%	+ 11.8%	+ 3.8%

1b-4 solution

a.

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

- $Mean = \frac{-0.043+\cdots+0.038}{4} = 0.09925 = 9.93\%$

b.

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}; S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

- $S^2 = \frac{(-0.043-0.09925)^2+\cdots+(0.038-0.09925)^2}{4-1} = 0.01949$
- $S = \sqrt{0.01949} = 0.1396 = 13.96\%$

1b-5

5. Using the data in the table [5],
   a. What was the average annual return of Microsoft stock from 2002-2014?
   b. What was the annual volatility for Microsoft stock from 2002-2014?

Review Only

Click the icon to see the Worked Solution (Formula Solution).

a. What was the average annual return of Microsoft stock from 2002-2014?

The average annual return is _____%. (Round to two decimal places.)

b. What was the annual volatility for Microsoft stock from 2002-2014?

The annual volatility of returns is _____%. (Round to two decimal places.)

1b-5 (Cont.)

5: Data Table

(*Click on the icon located on the top-right corner of the data table below in order to copy its contents into a spreadsheet.*)

Realized Return for the S&P 500, Microsoft, and Treasury Bills, 2002-2014

Year End	S&P 500 Index	Dividends Paid*	S&P 500 Realized Return	Microsoft Realized Return	1-Month T-Bill Return
2001	1148.08				
2002	879.82	14.53	−22.1%	−22.0%	1.6%
2003	1111.92	20.80	28.7%	6.8%	1.0%
2004	1211.92	20.98	10.9%	8.9%	1.2%
2005	1248.29	23.15	4.9%	−0.9%	3.0%
2006	1418.30	27.16	15.8%	15.8%	4.8%
2007	1468.36	27.86	5.5%	20.8%	4.7%
2008	903.25	21.85	−37.0%	−44.4%	1.5%
2009	1115.10	27.19	26.5%	60.5%	0.1%
2010	1257.64	25.44	15.1%	−6.5%	0.1%
2011	1257.60	26.59	2.1%	−4.5%	0.0%
2012	1426.19	32.67	16.0%	5.8%	0.1%
2013	1848.36	39.75	32.4%	44.2%	0.0%
2014	2058.90	42.47	13.7%	27.5%	0.0%

*Total dividends paid by the 500 stocks in the portfolio, based on the number of shares of each stock in the index, adjusted until the end of the year, assuming they were reinvested when paid.

Source: Standard & Poor's, Microsoft and U.S. Treasury Data

1b-5 solution

a.

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

- $Mean = \frac{-0.22+\cdots+0.275}{13} = 0.086154 = 8.62\%$

b.

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}; S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

- $S^2 = \frac{(-0.22-0.0862)^2+\cdots+(0.275-0.0862)^2}{13-1} = 0.73044$
- $S = \sqrt{0.073044} = 0.2703 = 27.03\%$

Recap: measures of variation

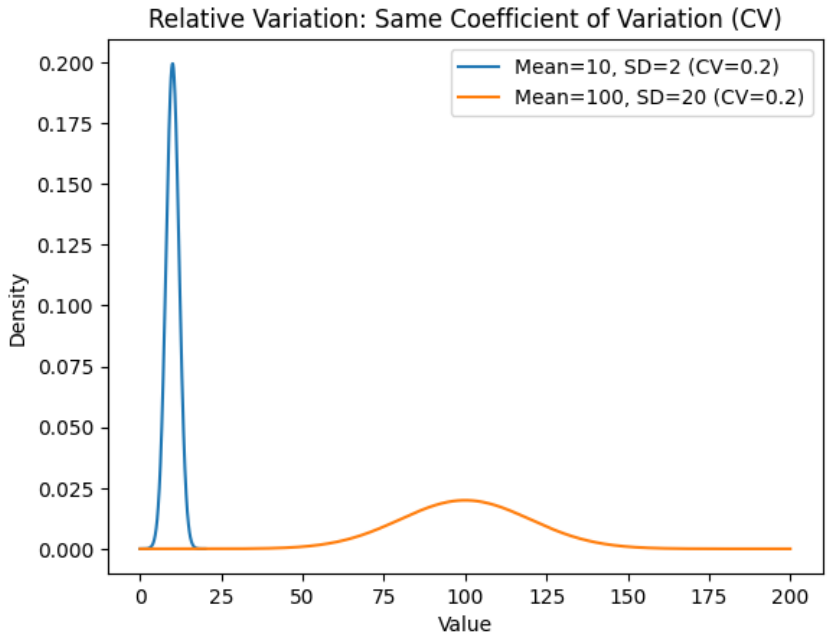
- So far, we have discussed spread using measures such as range, variance, and standard deviation. These measures describe **absolute** spread.
- However, when two datasets have different means, absolute spread alone may not give the full picture, because it does not allow for a fair comparison of variability across different scales.
- In that case, the coefficient of variation, or CV, becomes useful because it measures spread **relative to** the mean.

Recap: measures of variation

- Formula:

$$CV = \left(\frac{S}{\bar{X}}\right) \times 100\%$$

- Rule:
  - Larger CV → data are more spread out relative to the mean
  - Smaller CV → data are more concentrated around the mean



Recap: measures of variation

- Stock A:
  - Average price last year = $50
  - Standard deviation = $5

$$CV_A = \left(\frac{S}{\bar{X}}\right) \cdot 100\% = \frac{\$5}{\$50} \cdot 100\% = 10\%$$

- Stock B:
  - Average price last year = $100
  - Standard deviation = $5

$$CV_B = \left(\frac{S}{\bar{X}}\right) \cdot 100\% = \frac{\$5}{\$100} \cdot 100\% = 5\%$$

Both stocks have the same standard deviation, but stock B is less variable relative to its price

1a-5

6. The market capitalizations of 20 companies, in billions of dollars, are listed to the right. Complete parts (a) and (b).

229.9	178.3	223.8	63.8	66.7
75.1	235.1	173.1	107.3	58.6
220.2	240.2	318.1	179.1	17.2
16.4	512.7	466.2	22.7	553.8

a. Calculate the mean and standard deviation of the market capitalization for this population of 20 companies.

The mean is = _____. (Round to three decimal places as needed.)

The standard deviation is = _____.
(Round to three decimal places as needed.)

b. Interpret the parameters calculated in (a). Assume that for market capitalizations, a standard deviation is large as a portion of the mean if it is greater than 25% of the mean. Choose the correct answer below.

- ○ **A.** The standard deviation is large as a portion of the mean. This indicates that the values are relatively concentrated.
- ○ **B.** The standard deviation is small as a portion of the mean. This indicates that the values are relatively concentrated.
- ○ **C.** The standard deviation is small as a portion of the mean. This indicates that the values are relatively spread out.
- ○ **D.** The standard deviation is large as a portion of the mean. This indicates that the values are relatively spread out.

1a-5 solution

a.

- $Mean = \frac{229.9+178.3+\cdots+553.8}{20} = 197.915$
- $\sigma^2 = \frac{\sum_{i=1}^{n}(X_i-\mu)^2}{n} = \frac{(229.9-197.915)^2+\cdots+(553.8-197.915)^2}{20} = 24654.67$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(X_i-\mu)^2}{n}} = \sqrt{24654.67} = 157.018$$

1a-5 solution

b.

- **D.** The standard deviation is large as a portion of the mean. This indicates that the values are relatively large spread out.
- $CV = \left(\frac{S}{\bar{X}}\right) \times 100\% = \left(\frac{157.018}{197.915}\right) \times 100\% = 78.34\% > 25\%$

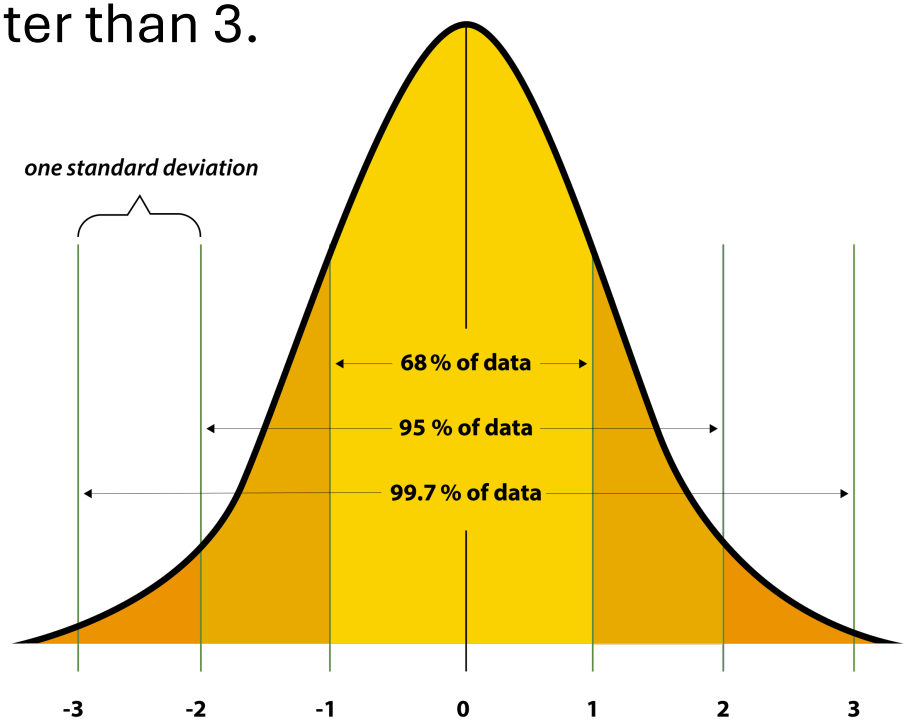
Recap: measures of variation

- So far, we have looked at measures of variation that describe the spread of a whole dataset. But sometimes, we are not interested in the whole dataset. Instead, we want to know where **one particular value** stands.
- For example, is it above the mean, below the mean, or far away from the mean?
- A **z-score** is the number of standard deviations a data value is from the mean.

$$Z = \frac{X - \bar{X}}{S}$$

Recap: measures of variation

- A value is often considered a potential **outlier** if its z-score is less than -3 or greater than 3.



1a-3

3. A bank manager records the following waiting times, in minutes, of a sample of clients.

   4.19 5.53 3.04 2.32 0.40 5.14

   a. Compute the mean and median.
   b. Compute the variance, standard deviation, range, coefficient of variation, and Z scores. Are there any outliers?
   c. Are the data skewed? If so, how?
   d. The bank manager claims that each client can expect to wait "almost certainly less than five minutes." Based on the given data, is this accurate?

a. The mean wait time is _____ minutes.
(Round to two decimal places as needed.)

The median wait time is _____ minutes.
(Round to two decimal places as needed.)

b. The variance is _____ minutes.
(Round to three decimal places as needed.)

The standard deviation is _____ minutes.
(Round to three decimal places as needed.)

The range is _____ minutes.
(Type an integer or a decimal. Do not round.)

The coefficient of variation is _____%.
(Round to two decimal places as needed.)

1a-3 solution

a.

- $Mean = \frac{4.19+5.53+\cdots+5.14}{6} = 3.44$
- $0.40 \quad 2.32 \quad 3.04 \quad 4.19 \quad 5.14 \quad 5.53$
- $Median = \frac{3.04+4.19}{2} = 3.62$

Excel methods

$= \text{AVERAGE(DataRange)}$
$= MEDIAN(DataRange)$

1a-3 solution

b.

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}; S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

$$Variance = 3.695; Standard\ deviation = \sqrt{3.695} = 1.922$$

Excel methods

- Step-by-step:
  $= SUM(SquaredDeviationsRange)/COUNT(DataRange) - 1$
- Built-in function:
  $= VAR.S(DataRange)$ $= STDEV.S(DataRange)$

1a-3 solution

b.

- $0.40 \quad 2.32 \quad 3.04 \quad 4.19 \quad 5.14 \quad 5.53$
- $Range = 5.53 - 0.40 = 5.13$
- $CV = \left(\frac{S}{\bar{X}}\right) \times 100\% = \left(\frac{1.922}{3.44}\right) \times 100\% = 55.87\%$

1a-3 (Cont.)

Compute the Z scores.

Data	Z Score	
4.19	_____	(Round to two decimal places as needed.)
5.53	_____	(Round to two decimal places as needed.)
3.04	_____	(Round to two decimal places as needed.)
2.32	_____	(Round to two decimal places as needed.)
0.40	_____	(Round to two decimal places as needed.)
5.14	_____	(Round to two decimal places as needed.)

Are there any outliers?

- ○ **A.** Yes, since 0.40 is far away from the median.
- ○ **B.** Yes, since 0.40 is far away from the mean.
- ○ **C.** No, since this set of data does not have a mode.
- ○ **D.** No, since there are no Z scores that are less than -3 or greater than 3.

1a-3 (Cont.) solution

b.

Data	Z Score
4.19	0.39
5.53	1.09
3.04	-0.21
2.32	-0.58
0.40	-1.58
5.14	0.89

$$Z = \frac{X - \bar{X}}{S}$$

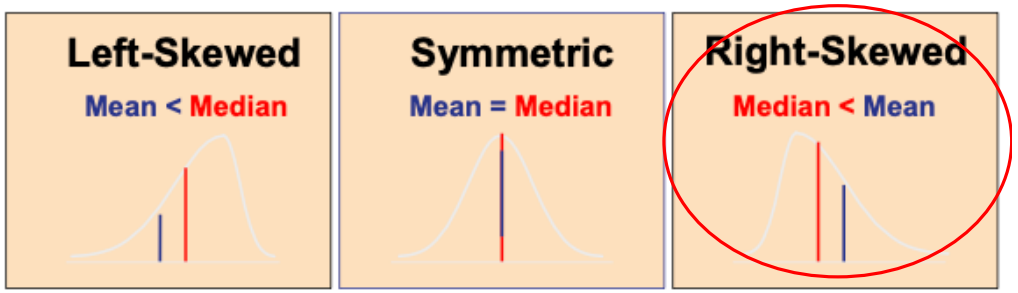
D. No outliers, since there are no Z scores that are less than 3 or greater than 3.

Recap: shape of a distribution

- While variation tells us the amount of spread, shape tells us how the data are distributed across the values.
- Shape helps answer questions like: “Is it symmetric or skewed?” and “Are the tails heavy or light?”
- Two key features of shape are
  - Skewness – the degree of asymmetry
  - Kurtosis – the heaviness of tails / peakedness

Recap: shape of a distribution

- Skewness
  - Left-skewed: mean < median (long tail on the left)
  - Symmetric: mean = median
  - Right-skewed: mean > median (long tail on the right)
- Kurtosis
  - Leptokurtic: sharper peak, heavier tails
  - Mesokurtic: normal bell shape
  - Platykurtic: flatter peak, lighter tail



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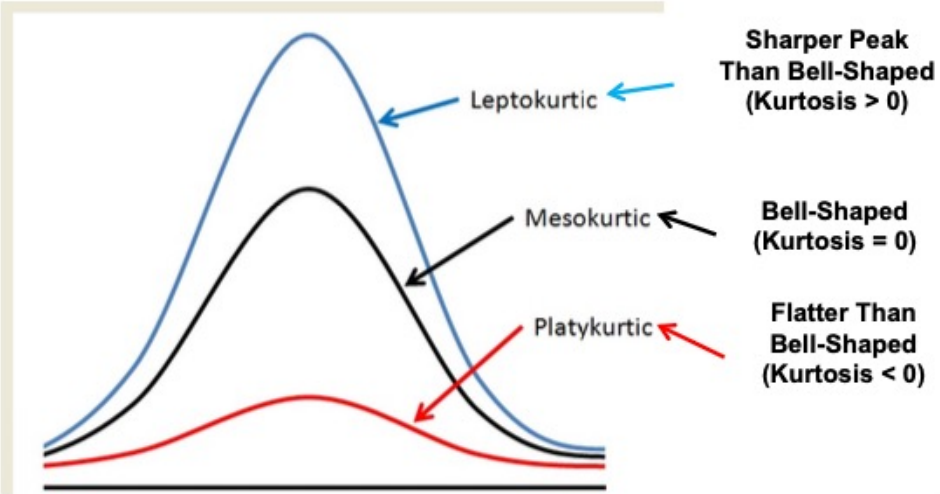
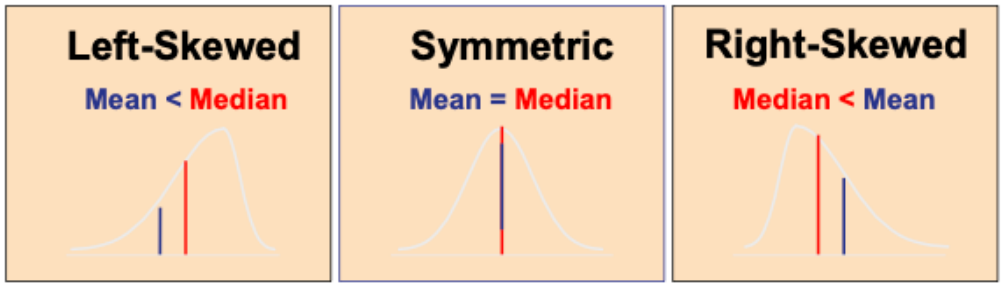




Recap: using Excel functions

House Prices		Descriptive Statistics		
$ 2,000,000		Mean	$ 600,000	=AVERAGE(A2:A6)
$ 500,000		Standard Error	$ 357,770.88	=D6/SQRT(D14)
$ 300,000		Median	$ 300,000	=MEDIAN(A2:A6)
$ 100,000		Mode	$ 100,000.00	=MODE(A2:A6)
$ 100,000		Standard Deviation	$ 800,000	=STDEV(A2:A6)
		Sample Variance	640,000,000,000	=VAR(A2:A6)
		Kurtosis	4.1301	=KURT(A2:A6)
		Skewness	2.0068	=SKEW(A2:A6)
		Range	$ 1,900,000	=D12 - D11
		Minimum	$ 100,000	=MIN(A2:A6)
		Maximum	$ 2,000,000	=MAX(A2:A6)
		Sum	$ 3,000,000	=SUM(A2:A6)
		Count	5	=COUNT(A2:A6)

1a-3 (Cont.)

c. Are the data skewed?

- **A.** Yes, they are right-skewed.
- **B.** Yes, they are left-skewed.
- **C.** No, they are not skewed.

d. Is the bank manager's claim accurate?

- **A.** No, since the two times that are greater than five minutes have Z scores less than 3.
- **B.** Yes, since the given times that are greater than five minutes are outliers.
- **C.** Yes, since the two times that are greater than five minutes have Z scores less than 3.
- **D.** It cannot be determined since "almost certainly" is too vague to be analyzed.

d. The bank manager claims that each client can expect to wait "almost certainly less than five minutes." Based on the given data, is this accurate?

1a-3 (Cont.) solution

- **B.** Yes, they are left-skewed.
  - $Mean < Median$ $(3.44 < 3.62)$
- **A.** No, since the two times that are greater than five minutes have Z scores less than 3.
  - Since the two waiting times above 5 minutes **are not outliers**, waits longer than 5 minutes are still plausible.

$Z - Score < 3$

Data	Z Score
4.19	0.39
5.53	1.09
3.04	-0.21
2.32	-0.58
0.40	-1.58
5.14	0.89

Recap: probability and expected return

- So far, we have used statistics to describe **observed data** in terms of center, spread, and shape.
- In many real-world situations, especially in finance, outcomes are **uncertain** and may occur with different probabilities.
- The probabilities must satisfy:

$$0 \leq P(R_i) \leq 1$$

$$\sum P(R_i) = 1$$

Recap: probability and expected return

- The **expected return** is the probability-weighted average of all possible returns.
- Formula:

$$E(R) = \sum P(R_i) \times R_i$$

Expected return vs. Mean

- Mean (arithmetic average): each observed return gets **equal** weight.
- Expected return: each possible return gets weighted by its **probability**.

Recap: probability and expected return

- To measure risk, we examine how far possible returns deviate from the expected return.
- <u>Two common measures</u> of risk are: (i) variance of return; (ii) standard deviation of return
- Formula:

$$Var(R) = \sum P(R_i)(R_i - E(R))^2$$

$$\sigma_R = \sqrt{\sum P(R_i)(R_i - E(R))^2}$$

1b-1

1. The figure in the popup window, [1], shows the one-year return distribution for RCS stock. Calculate:
   a. The expected return.
   b. The standard deviation of the return.

   Note: Make sure to round all intermediate calculations to at least five decimal places.

Review Only

Click the icon to see the Worked Solution (Formula Solution).

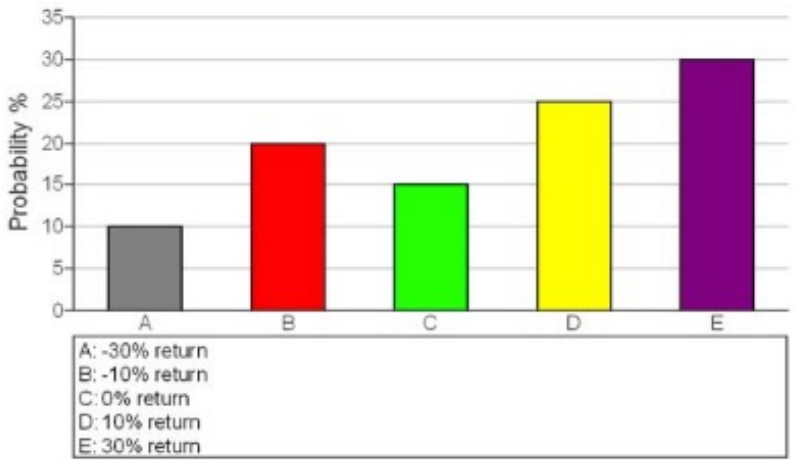
a. The expected return.

The expected return is _____%. (Round to two decimal places.)

b. The standard deviation of the return.

The standard deviation is _____%. (Round to two decimal places.)

1: Graph/Chart



1b-1 solution

a.

$$E(R) = \sum P(R_i) \times R_i$$

- $E(R) = (0.1) \times (-0.3) + \cdots + (0.3) \times (0.3) = 0.065 = 6.50\%$

b.

$$Var(R) = \sum P(R_i)(R_i - E(R))^2 \qquad \sigma_R = \sqrt{\sum P(R_i)(R_i - E(R))^2}$$

- $Var(R) = 0.1 \times (-0.3 - 0.065)^2 + \cdots + 0.3 \times (0.3 - 0.065)^2 = 0.036275$
- $\sigma_R = \sqrt{0.036275} = 0.1905 = 19.05\%$

1b-2

2. The table here, [2], shows the one-year return distribution of Startup Inc. Calculate the:
   a. Expected return.
   b. The standard deviation of the return.

Review Only

Click the icon to see the Worked Solution (Formula Solution).

a. Expected return:

The expected return is _____%. (Round to one decimal place.)

b. The standard deviation of the return.

The standard deviation is _____%. (Round to one decimal place.)

2: Data Table

(*Click on the icon located on the top-right corner of the data table below in order to copy its contents into a spreadsheet.*)

Probability	40%	20%	20%	10%	10%
Return	− 100%	− 85%	− 40%	− 25%	1,000%

1b-2 solution

a.

$$E(R) = \sum P(R_i) \times R_i$$

- $E(R) = (0.4)(-1) + \cdots + (0.1)(10) = 0.325 = 32.5\%$

b.

$$Var(R) = \sum P(R_i)(R_i - E(R))^2 \qquad \sigma_R = \sqrt{\sum P(R_i)(R_i - E(R))^2}$$

- $\sigma_R = \sqrt{(0.4)(-1 - 0.325)^2 + \cdots + (0.1)(0.1 - 0.325)^2} = 3.237 = 323.7\%$

Recap: financial returns

- So far, we have compared the arithmetic mean and the geometric mean. And the geometric mean is appropriate for connected changes over time. In finance, these connected changes are measured by **returns**.
- A one-period **net return** measures the percentage change in price from $t-1$ to t.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

- Equivalently, the **gross return** is:

$$1 + R_t = \frac{P_t}{P_{t-1}}$$

Recap: financial returns

- Over multiple periods, returns are linked through compounding:

$$1+R_t^{(k)} = \frac{P_t}{P_{t-k}} = \frac{P_t}{P_{t-1}}\frac{P_{t-1}}{P_{t-2}}\cdots\frac{P_{t-k+1}}{P_{t-k}}$$
$$=(1+R_t)\cdots(1+R_{t-k+1})$$

$$R_t^{(k)} = \frac{P_t - P_{t-k}}{P_{t-k}} = \frac{P_t}{P_{t-k}} - 1$$
$$= \frac{P_t}{P_{t-1}}\frac{P_{t-1}}{P_{t-2}}\cdots\frac{P_{t-k+1}}{P_{t-k}} - 1$$
$$= (1+R_t)\cdots(1+R_{t-k+1}) - 1$$

where t denotes the current time period, and k denotes the number of periods over which the return is measured.

- Therefore, when summarizing investment performance over time, the geometric mean is the natural choice.

Recap: financial returns

The one-period log return is

$$r_t = \ln(1+R_t) = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1}$$

- The main advantage of using log returns is
  - that the multi-period LOG return is simply the sum of one-period LOG returns

$$\begin{aligned} r_t^{(k)} &= \ln\left\{1+R_t^{(k)}\right\} = \ln\left\{(1+R_t)\cdots(1+R_{t-k+1})\right\} \\ &= \ln(1+R_t)+\cdots+\ln(1+R_{t-k+1}) \\ &= r_t + r_{t-1} + \cdots + r_{t-k+1} \end{aligned}$$

- *so that* $r_t^{(k)} = \sum_{j=0}^{k-1} r_{t-j}$

APQ-1

1. The prices of a stock are given in the table on the right.
   (a) What is R_2?
   (b) What is $R_4^{(3)}$?
   (c) What is r_3?

t	P_t
1	51
2	56
3	53
4	58

APQ-1solution

(a) $R_2 = \frac{P_2}{P_1} - 1 = \frac{56}{51} - 1 = 9.80\%$

(b) $R_4^{(3)} = (1+R_4)(1+R_3)(1+R_2) - 1 = \frac{P_4}{P_3} \cdot \frac{P_3}{P_2} \cdot \frac{P_2}{P_1} - 1$

$= \frac{58}{53} \cdot \frac{53}{56} \cdot \frac{56}{51} - 1 = \frac{58}{51} - 1 = 13.73\%$

(c) $r_3 = \ln \frac{P_3}{P_2} = \ln \frac{53}{56} = -5.51\%$

APQ-2

2. The prices of a stock are given in the table on the right. Calculate

(a) R_2, R_3 and R_4.

(b) $R_3^{(2)}$, $R_4^{(3)}$ and $R_5^{(3)}$.

(c) r_3 and r_5.

(d) $r_3^{(2)}$, $r_4^{(3)}$ and $r_5^{(3)}$.

t	P_t
1	72
2	74
3	69
4	70
5	71

APQ-2 solution

(a) $R_2 = \frac{P_2}{P_1} - 1 = \frac{74}{72} - 1 = 2.78\%$

$R_3 = \frac{P_3}{P_2} - 1 = \frac{69}{74} - 1 = -6.76\%$

$R_4 = \frac{P_4}{P_3} - 1 = \frac{70}{69} - 1 = 1.45\%$

(b) $R_3^{(2)} = (1+R_3)(1+R_2) - 1 = \frac{P_3}{P_2} \cdot \frac{P_2}{P_1} - 1 = \frac{69}{74} \times \frac{74}{72} - 1 = \frac{69}{72} - 1 = -4.17\%$

$R_4^{(3)} = (1+R_4)(1+R_3)(1+R_2) - 1 = \frac{P_4}{P_3} \cdot \frac{P_3}{P_2} \cdot \frac{P_2}{P_1} - 1 = \frac{70}{72} - 1 = -2.78\%$

$R_5^{(3)} = (1+R_5)(1+R_4)(1+R_3) - 1 = \frac{P_5}{P_4} \cdot \frac{P_4}{P_3} \cdot \frac{P_3}{P_2} - 1 = \frac{P_5}{P_2} - 1 = \frac{71}{74} - 1 = -4.05\%$

(c) $r_3 = \ln \frac{P_3}{P_2} = \ln \frac{69}{74} = -7.00\%$

$r_5 = \ln \frac{P_5}{P_4} = \ln \frac{71}{70} = 1.42\%$

(d) $r_3^{(2)} = \ln\left[1 + R_3^{(2)}\right] = \ln\left[\frac{P_3}{P_2} \cdot \frac{P_2}{P_1}\right] = \ln \frac{69}{72} = -4.26\%$

$r_4^{(3)} = \ln\left[1 + R_4^{(3)}\right] = \ln\left[\frac{P_4}{P_3} \cdot \frac{P_3}{P_2} \cdot \frac{P_2}{P_1}\right] = \ln \frac{70}{72} = -2.82\%$

$r_5^{(3)} = \ln\left[1 + R_5^{(3)}\right] = \ln\left[\frac{P_5}{P_4} \cdot \frac{P_4}{P_3} \cdot \frac{P_3}{P_2}\right] = \ln \frac{71}{74} = -4.14\%$

APQ-3

3 The prices of a stock in months 1-5 are given below. Please compute the corresponding single-period, multi-period net, gross and logarithmic returns and fill in the blanks in the table.

t	P_t	R_t	$1+R_t$	$1+R_t^{(3)}$	r_t	$r_t^{(3)}$
1	200	×	×	×	×	×
2	192			×		×
3	198			×		×
4	206					
5	210					

APQ-3 solution

t	P_t	R_t	$1+R_t$	$1+R_t^{(3)}$	r_t	$r_t^{(3)}$
1	200	×	×	×	×	×
2	192	-0.0400	0.9600	×	-0.0408	×
3	198	0.0313	1.0313	×	0.0308	×
4	206	0.0404	1.0404	1.03	0.0396	0.0296
5	210	0.0194	1.0194	1.09375	0.0192	0.0896

APQ-4

4 True or False questions

(1) The k-period log return is the product of k terms of single-period log returns.

(2) Log returns are also called continuously-compounded returns.

APQ-4 solution

- False
- True